

Nmap Scan Types

Cat.	Flag	Scan Name	Description / How it works	Response interpretation	Root Required	Speed	Noise Level	When to use
CORE TCP / UDP PORT SCANS								
1	-sS	TCP SYN Scan (Stealth / Half-Open)	Sends a SYN packet (step 1 of the handshake), then judges port state from the response — without ever completing the handshake. On SYN-ACK, Nmap sends RST to tear down. Default scan when running as root.	SYN-ACK → open RST → closed No response / ICMP unreachable → filtered	Yes	Fast	Low–Med	Default TCP workhorse. Use when you have privileges and want speed + slightly lower noise. Modern IDS detects easily, so "stealth" is mostly historical.
2	-sT	TCP Connect Scan	Asks the OS to make a full TCP connection via the standard connect() system call. The kernel performs the entire 3-way handshake. No raw socket access needed.	Connection succeeds → open RST (refused) → closed Timeout / unreachable → filtered	No	Medium	High	When you don't have root (restricted shells, some Windows setups, containers). Nmap falls back to this automatically for non-root users. Target apps usually log completed connections.
3	-sU	UDP Scan	Sends a UDP packet to each port. Since UDP is connectionless, Nmap infers state from weak signals. Often sends protocol-specific payloads (DNS, SNMP, NTP) to coax responses from services.	UDP response → open ICMP type 3, code 3 → closed ICMP code 1,2,9,10,13 → filtered No response → open filtered	Yes	Very Slow	Medium	Finding UDP services — DNS (53), SNMP (161), NTP (123), DHCP (67/68), VPNs. Often overlooked = rich attack surface. Use --top-ports 100 instead of full sweep.
ENUMERATION & FINGERPRINTING								
4	-sV	Service / Version Detection	After finding open ports, probes them to identify the exact service and version. Goes beyond "port 22 open" to "OpenSSH 8.2p1 Ubuntu". Tunable via --version-intensity 0-9.	Banner match → service + version Soft match → service only No match → fingerprint shown for submission	No	Medium	High	Critical for matching vulnerabilities to CVEs. Always run after initial port discovery — knowing "Apache 2.4.49" vs "open port 80" is the difference between finding RCE and missing it.
5	-sC	Default Script Scan	Runs Nmap's default NSE (Nmap Scripting Engine) scripts against open ports. Equivalent to --script=default. Pulls banners, enumerates SMB shares, grabs SSL certs, checks common misconfigs.	Script output → service details, shares, certs, vulns - Output appears under each port in scan results	No*	Medium	High	Use after identifying open ports for quick wins. Combined with -sV in the -A shortcut. Some scripts need root; safe-category scripts run by default.
6	-O	OS Detection	Fingerprints the target OS by analyzing TCP/IP stack quirks — window sizes, TTL values, option ordering, ICMP behavior. Compares against Nmap's signature database.	OS match → vendor, family, generation, kernel Aggressive guess → multiple possibilities ranked No match → fingerprint shown	Yes	Medium	Medium	Tailor exploits to the target (Windows vs Linux, kernel hints). Less reliable behind NAT/firewalls or against modern stacks. Needs at least 1 open + 1 closed port for best results.
7	-A	Aggressive Scan	All-in-one shortcut: -sV -sC -O --traceroute. Runs version detection, default scripts, OS fingerprinting, and traceroute in a single flag. Information-rich but loud.	- Combined output of all included scans - Service versions + script results + OS guess + route	Yes	Slow	Very High	The "tell me everything" command. Use during authorized engagements where stealth doesn't matter. Never use when evading detection is a goal — it lights up every IDS sensor.
HOST DISCOVERY								
8	-Pn	Skip Host Discovery (No Ping)	By default Nmap pings first and skips "down" hosts. -Pn forces a full port scan regardless of ping response. Treats every target as alive.	- Skips ping check entirely - Proceeds straight to port scanning - Slower if host is truly offline	No	Varies	N/A	Essential when targets block ICMP (most modern firewalls do). Junior mistake to avoid: reporting "host is down" when really ICMP is just filtered. Add to almost every external scan.
9	-sn	Ping Sweep (No Port Scan)	Discovers which hosts are alive on a network without scanning any ports. Uses ICMP echo, TCP SYN to 443, TCP ACK to 80, and ICMP timestamp by default (as root).	Any response → host up No response → host down (or filtered) - Outputs list of live IPs only	Optional	Very Fast	Low	Map a subnet before deeper scanning. Quick way to identify live hosts in a /24 or larger range. First step in most internal engagements: nmap -sn 10.0.0.0/24.
STEALTH & FIREWALL ANALYSIS								
10	-sA	TCP ACK Scan	Sends TCP packets with only the ACK flag set. Doesn't identify open/closed ports — instead maps firewall rulesets. Helps determine if a firewall is stateful and which ports it filters.	RST received → unfiltered (firewall lets it through) No response / ICMP unreachable → filtered (stateful firewall blocking)	Yes	Fast	Low	Understanding network architecture, not port states. Pair with -sS results to distinguish "firewall blocks all" vs "firewall is stateful". Useful before crafting evasion strategies.

Advanced / situational (not pictured): -sN Null, -sF FIN, -sX Xmas (stealth via odd flags, fail against Windows) · -sI <zombie> Idle scan (truly stealthy, hard to set up) · -PR ARP ping (auto-used on LAN) · -PS/-PA/-PU/-PE alternate ping methods when ICMP is blocked.